

CFD Report: Flow and Heat Transfer in a 2D Pipe Bend using ANSYS Fluent

1. Objective

The objective of this project is to perform a Computational Fluid Dynamics (CFD) analysis of fluid flow and heat transfer in a 2D pipe bend geometry using ANSYS Fluent. The study focuses on understanding velocity distribution and temperature mixing.

2. Problem Description

Two inlets with different velocities and temperatures are considered:

- Inlet 1: 0.4 m/s, 20°C

- Inlet 2: 1.2 m/s, 40°C

Outlet: Pressure outlet

Fluid properties:

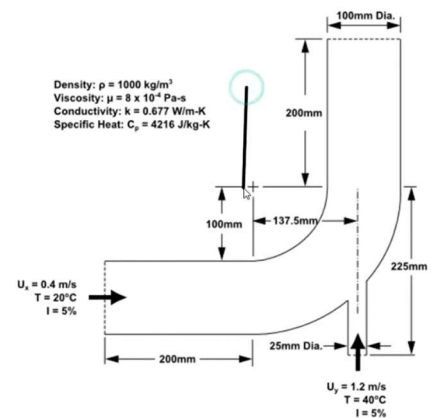
Density: 1000 kg/m³

Viscosity: 8×10^{-4} Pa·s

Thermal Conductivity: 0.677 W/m·K

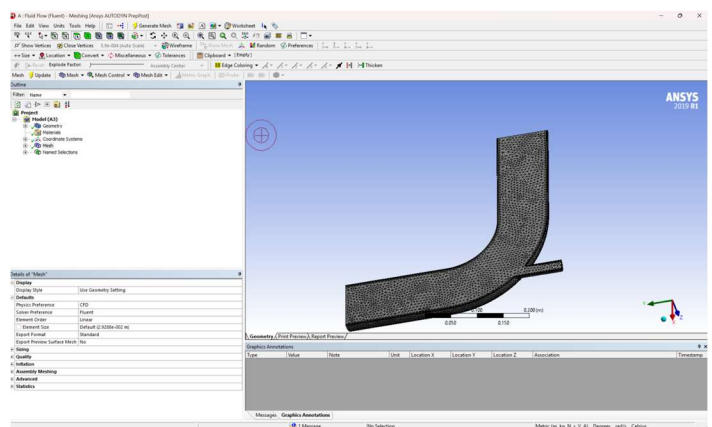
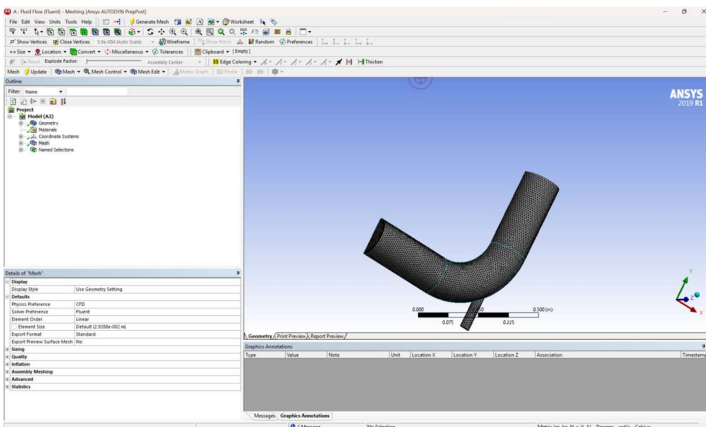
Specific Heat: 4216 J/kg·K

Figure 1.1: Problem Specification



3. Geometry and Mesh

The geometry was created based on given dimensions. A fine mesh using triangular elements was generated with refinement near bends and junctions.



4. Numerical Setup

Solver: Pressure-based

Type: Steady-state

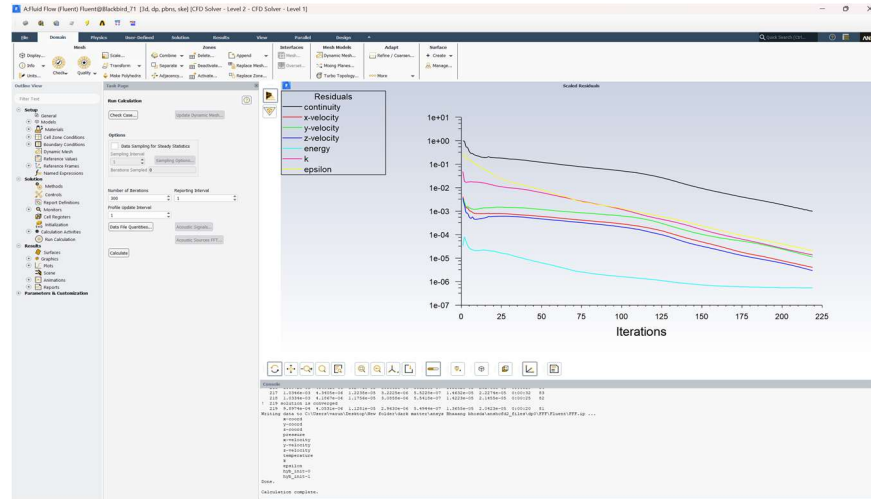
Model: k-epsilon turbulence model

Energy Equation: Enabled

Boundary Conditions: Velocity inlets and pressure outlet

5. Solution and Convergence

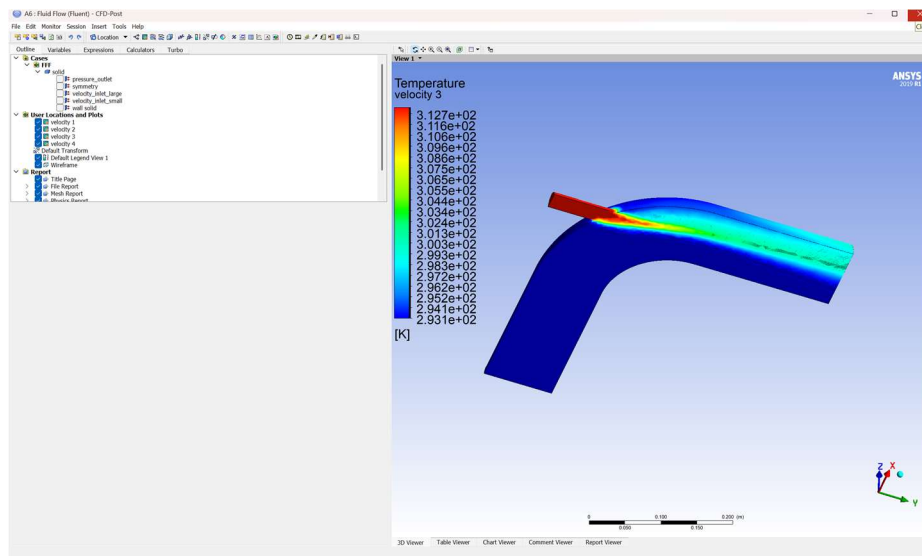
The simulation was run for 300 iterations. Residuals showed steady convergence.



6. Results and Discussion

Velocity distribution shows dominance of higher velocity inlet.

Temperature contours show gradual mixing of hot and cold streams.



7. Conclusion

The simulation successfully captured flow behavior and heat transfer. Mixing is influenced by geometry and inlet conditions.

8. Future Scope

Mesh independence study, transient analysis, and turbulence model comparison can be explored.